AIAA 3201 Project Release

Project Brief

- Four advanced topics for you to choose, please check Canvas documents for details
  - Video Super-Resolution
  - Few-Shot Industrial Visual Anomaly Detection
  - Monocular Metric Depth Estimation
  - Generative Sparse-View 3D Reconstruction

Project Requirement

- Team formation
  - **Please find your teammate as soon as possible!**

- Computational resources
  - You may apply the online computing credits from TAs, when you try to run deep learning models
  - Note the credit is limited for each student, please do not waste

- Project timeline
  - **Week 11 & 12 (18th and 25th Nov.)**, present your project progress in class.
  - **11th December 11.59PM**, project submission in Canvas deadline

Project Oral Presentation

- **At least Part 1** of the project should be completed, preferably also complete Part 2

- As L01 and L02 have unbalanced student number, **part of the teams need to go L02 for presentation**

- Each team submit a **4-min video recording** of your presentation

- **No need to worry!** The presentation contributes **very little (5%)** to your final grades.

We would like to promote idea exchange

- You need to give introduction on your **basic methods, experiments and the plan for your next step**

- **Evaluate:**
  - **Peer review: every student can give a rating for others**
  - **Discussion matters! Each team will be the Discussant of another team with the same topic, which is expected to prepare at least one meaningful question.**
  - **TA & Me grading (Implementation, Experiments, Demos, etc.)**

Project Final Submission

- We follow **standard** research publication release process
  - **(Mandatory)** Release your code to the public by GitHub
  - **(Optional)** A demo video or a project page
  - **(Optional)** Upload your report to Preprint

- Zip **all the above files and the Github link**, submit them in Canvas, by **11th December 11.59PM**

Project Report

- **Abstract & Introduction**: Background, motivation and solution overview. Explicitly include your GitHub repository link at the end of abstract.
- **Related Work (IMPORTANT)**: You must review and cite **at least all papers** listed in the Reference Section. Expanding beyond this list is encouraged.
- **Method**: Explain the technical roadmaps in your own words.
- **Experiments**: Present quantitative results (Tables) and qualitative analysis (Figures).
- **Conclusion**: Summary, limitations, and future work.

Q & A



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